

Problem Set 6.1

1) $n \rightarrow$ red truck p $R \rightarrow$ # selected Red truck
 green truck $1-p$ $G \rightarrow$ # selected Green truck

a) $P_R(k) = \binom{n}{k} p^k (1-p)^{n-k}$ ✓ $E[R] = np$ ✓
 ; for $k=0, \dots, n$ $\text{var}(R) = np(1-p)$

b) $P(R=1) \text{ or } P(G=1)$
 ~~$P(R=1) \text{ or } P(G=1)$~~ X

[Two Cases] ~~ker~~

① 1st Package on R $\rightarrow (p)^1$
 rest Green $\rightarrow (1-p)^{n-1}$
 ② 1st Package on G $\rightarrow (1-p)^1$
 rest Red $\rightarrow (p)^{n-1}$

$P(\text{First Package \& only on Truck}) = p(1-p)^{n-1} + (1-p)p^{n-1}$

c) $P(R=1 \cup G=1) = \left[\binom{n}{1} p^1 (1-p)^{n-1} \right] + \left[\binom{n}{n-1} p^{n-1} (1-p)^1 \right]$

$B = \{R=1 \cup G=1\}$

$P(B) = \begin{cases} 1 & \text{if } n=1 \\ 2p(1-p) & \text{if } n=2 \\ \binom{n}{1} (1-p)^{n-1} p + \binom{n}{n-1} p^{n-1} (1-p) & \text{if } n=3, \dots \end{cases}$

✓ $2p(1-p) + (1-p)p$
 $\left\{ \begin{array}{l} p(1-p) + (1-p)p \end{array} \right\}$

$$d) E[R] = np$$

$$\begin{aligned} E[G] &= n - E[R] \\ &= n - np \\ &= n(1-p) \end{aligned}$$

$$\begin{aligned} E[R-G] &= E[R] - E[G] = np - n(1-p) \\ &= np - n + np \\ &= 2np - n \\ &= n(2p-1) \checkmark \end{aligned}$$

$$\text{var}(R) = np(1-p)$$

$$\text{var}(G) = \text{var}(n-R) = \text{var}(R) = np(1-p)$$

$$\begin{aligned} D &= R - G \\ &= R - (n-R) \\ &= 2R - n \end{aligned}$$

$$\begin{aligned} \text{var}(D) &= \text{var}(2R - n) = \\ &= 4\text{var}(R) \\ &= \underline{4np(1-p)} \checkmark \end{aligned}$$

e) Let $C \rightarrow$ event + first 2 packages on R

$$R = 2 + X_3 + X_4 + \dots + X_n$$

$$\begin{aligned} E[R|C] &= 2 + \cancel{E[R|C]} E[X_{3-n}] \\ &= 2 + E[X_3 + X_4 + \dots + X_n] \\ &= 2 + (n-2)E[X_i] \\ &= 2 + (n-2)p \\ &= 2 + \cancel{np-2p} (n-2)p \end{aligned}$$

$$\begin{aligned}
 \text{var}(R|C) &= 2 + \cancel{\text{var}(R|C)} \text{var}(X_3 + \dots + X_n) \\
 &= 2 + (n-2)^2 \text{var}(X_i) \\
 &= 2 + (n-2)^2 \cancel{\times} p(1-p) \\
 &= (n-2)^2 p(1-p)
 \end{aligned}$$

PMF?

$$p_{R|C}(r) = \binom{n-2}{r-2} p^{r-2} (1-p)^{n-r} \quad \text{for } r=2, \dots, n$$

2a) Fail $p = 1/4$ Pass $1-p = 3/4$

a) ~~PMF~~ $p_F(f) = \binom{n}{f} (1/4)^f (3/4)^{n-f}$

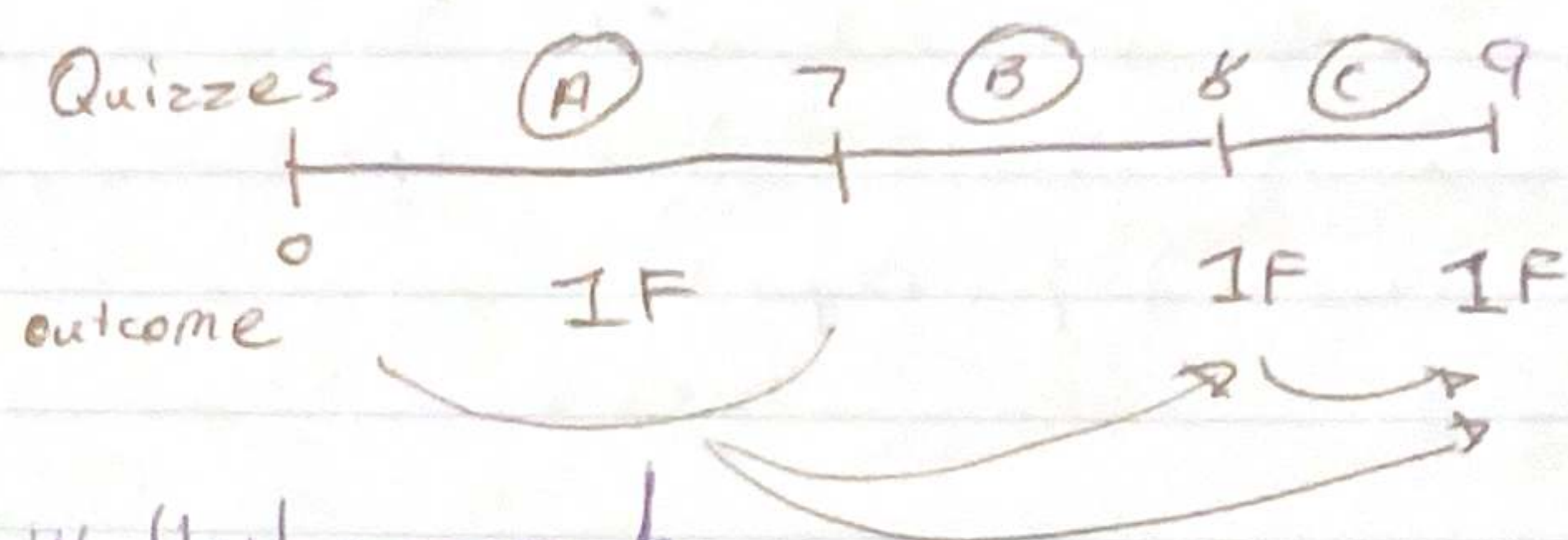
$$\begin{aligned}
 p_F(2) &= \binom{6}{2} (1/4)^2 (3/4)^{6-2} \\
 &= 0.2966 \\
 &= 29.66\%
 \end{aligned}$$

b) $E[\text{Quizzes to 3rd Fail}] = \text{Pascal order 3}$
 $= \frac{n}{p}$
 $= \frac{12}{(1/4)} = 12 \rightarrow$

$$E[\text{Fails}] = np = (12)(1/4) = 3$$

∴ $E[\text{Passes before 3rd Fail}] = 12 - 3$
 $= 9 \checkmark \rightarrow$

2c) $P(\text{2nd fail 8th quiz and 3rd fail 9th quiz})$



Walkthrough

$$P(A) = \binom{7}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^6 = 0.311462$$

$$P(B) = \frac{1}{4}$$

$$P(C) = \frac{1}{4}$$

$\left\{ \begin{array}{l} \text{Independent each time} \end{array} \right\}$

$$P(A \cap B \cap C) = \binom{7}{1} \left(\frac{3}{4}\right)^6 \left(\frac{1}{4}\right)^1 \cdot \left(\frac{1}{4}\right)^1 \left(\frac{1}{4}\right)^1 = \binom{7}{1} \left(\frac{3}{4}\right)^6 \left(\frac{1}{4}\right)^3$$

$\rightarrow$ multiply through events happen sequentially

d) A $\rightarrow$ prob fails two quizzes in a row
B $\rightarrow$ prob passes two quizzes in a row

$$P(A) = \left(\frac{1}{4}\right)^2$$

$$P(B) = \left(\frac{3}{4}\right)^2$$

$P(A \text{ before } B)$

Walkthrough

~~B~~ C $\rightarrow$ A before B

F $\rightarrow$ fail ~~S~~ P - pass

$$P(C) = P(\underline{FF} \cup \underline{PFF} \cup \underline{FPFF} \cup \underline{PFPFF} \cup \dots) = \left(\frac{1}{4}\right)^2 + \frac{3}{4} \left(\frac{1}{4}\right)^2 + \frac{1}{4} \cdot \frac{3}{4} \left(\frac{1}{4}\right)^2 + \dots$$

$$P(C) = \frac{P(A)}{\# \text{ events}} + P(B) \text{ of events}$$

$$P(C) = \left[\left(\frac{1}{4}\right)^2 + \frac{1}{4} \cdot \frac{3}{4} \left(\frac{1}{4}\right)^2 + \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \left(\frac{1}{4}\right)^2 + \dots \right] \\ + \left[\frac{3}{4} \left(\frac{1}{4}\right)^2 + \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \left(\frac{1}{4}\right)^2 + \dots \right]$$

alternat $\frac{1}{4}$ start and $\frac{3}{4}$ start

$$P(C) = \frac{\left(\frac{1}{4}\right)^2}{1 - \frac{1}{4} \cdot \frac{3}{4}} + \frac{\frac{3}{4} \left(\frac{1}{4}\right)^2}{1 - \frac{3}{4} \cdot \frac{1}{4}} = \\ = \frac{(1 + \frac{3}{4}) \left(\frac{1}{4}\right)^2}{(1 - \frac{1}{4} \cdot \frac{3}{4})} = \frac{7}{22}$$

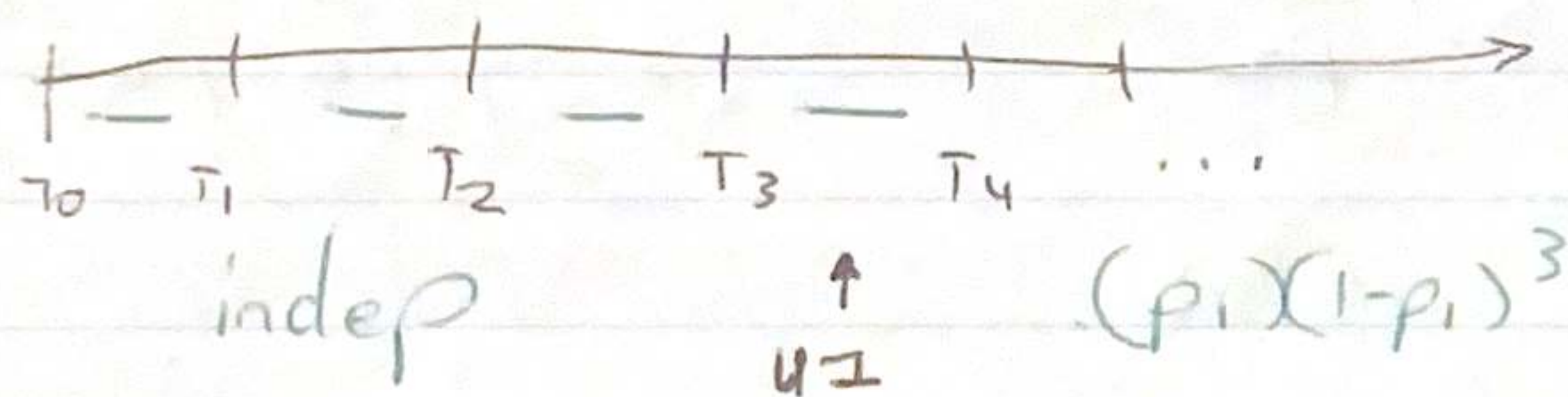
3a) 2 Users

Slots $\rightarrow$ idle $p_I = \frac{1}{6}$
 $\rightarrow$ busy $p_B = \frac{5}{6}$

$$p_{1/B} = \frac{2}{5}$$

$$p_{2/B} = \frac{3}{5}$$

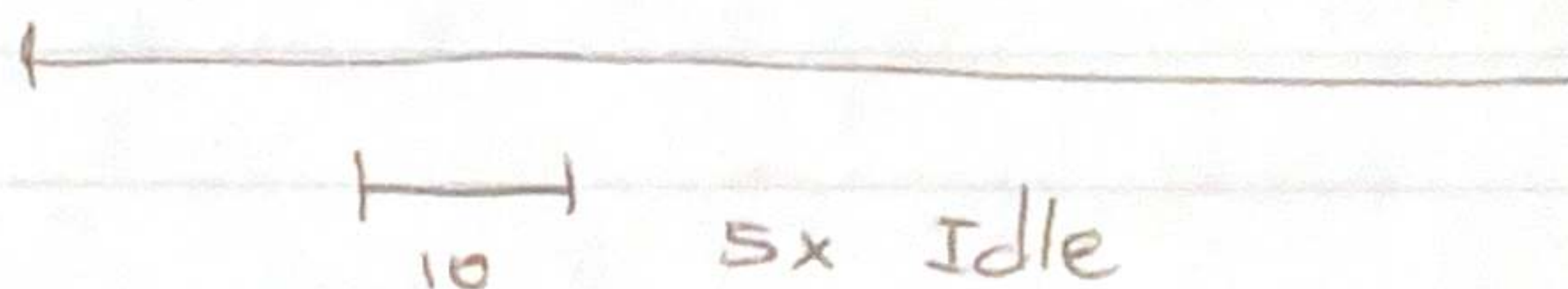
a) ~~4th First Task at~~ $P(T_1 \text{ first 4th slot}) = ?$



$$P_2 = P_{1/B} P_B = \left(\frac{2}{5}\right) \left(\frac{5}{6}\right) = \frac{1}{3} \quad \checkmark$$

$$P(\text{Task 1 for User 1 in Slot 4}) = p_1 (1-p_1)^3 \\ = \left(\frac{1}{3}\right) \left(1 - \frac{1}{3}\right)^3 \\ = \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^3 \\ = \frac{8}{81} \quad \checkmark$$

b)



$$P(\text{6th Idle Slot is Slot 12}) = P(A)$$

$$P(A) = P_I^5 P_B^6 \quad \times$$

→ Fresh Start → $\frac{B}{11\text{th Slot}} \quad \frac{I}{12\text{th Slot}}$

slot 1 to 10 irrelevant.

$$P(A) = P_B \cdot P_I = \left(\frac{5}{6}\right)\left(\frac{1}{6}\right) = \frac{5}{36}$$

c) ~~Let B = number of slots up to and including the 5th task from User I.~~

~~$$E[T] = \frac{1}{P_I} = \frac{1}{1/3} = 3 \text{ slots}$$~~

~~$$P_I = P_{I|B} \cdot P_B = \left(\frac{5}{6}\right)\left(\frac{2}{5}\right) = \frac{1}{3}$$~~

d) $P_{I|B} = \frac{2}{5}$

Number of ^{busy} Slots?
5x User I tasks

$$E[Y_k] = \frac{k}{p}$$

Pascal order 5

$$= \frac{5}{\left(\frac{2}{5}\right)} = \frac{25}{2}$$

How much time / discrete intervals?

⇒ Pascal.

c) Number of slots? $\rightarrow$ Pascal
 5th task $\rightarrow$ k order = 5
 User 1 $\rightarrow$ Condition on User 1

$$E[Y_k] = \frac{k}{p}$$

$$= \frac{5}{(1/3)}$$

$$= 15 \checkmark$$

$$k = 5$$

$$p_1 = p_{1|B} p_B = \left(\frac{2}{3}\right) \left(\frac{5}{6}\right)$$

$$= \frac{1}{3}$$

e) $p_{u2}(k) = \binom{15}{k} \left(\frac{2}{3}\right)^k \left(\frac{1}{3}\right)^{15-k}$

$$E[S] = 15 \left(\frac{2}{3}\right) = 10$$

$$\text{var}(S) = np(1-p) = 15 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)$$

$$= \frac{10}{3}$$

$\rightarrow$ Number of tasks? $\Rightarrow$ Pascal

$$P_{rk}(t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, t \geq k$$

$$P_B(t) = \binom{t-1}{5-1}$$

Num tasks User 2 until task 5 User 1
 portion Busy slots portion Busy slots

$$\text{Num User 2} = \# \text{ Busy slots} - 5$$

Num of Tasks $\rightarrow$ Pascal order 5 with param $p_{1|B} = \frac{2}{5}$

$$P_B(t) = \binom{t-1}{5-1} \left(\frac{2}{5}\right)^5 \left(1 - \frac{2}{5}\right)^{t-5}$$

$$= \binom{t-1}{4} \left(\frac{2}{5}\right)^5 \left(\frac{3}{5}\right)^{t-5}, t = 5, 6, \dots$$

$$E[B] = \frac{k}{p} \text{ where } T = B - 5$$

$$P_T(t) = P_B(t+5) = \binom{t+4}{4} \left(\frac{2}{5}\right)^5 \left(\frac{3}{5}\right)^t, t \geq 0$$

$$\begin{aligned}
 E[T] &= E[B-5] \\
 &= E[B] - 5 \\
 &= \frac{5}{(\frac{2}{5})} - 5 \\
 &= \frac{25}{2} - 5 \\
 &= \underline{7.5}
 \end{aligned}$$

$$\begin{aligned}
 \text{var}(T) &= \text{var}(B-5) = \text{var}(B) \\
 &= \frac{K(1-p)}{p^2} \\
 &= \frac{5(1-\frac{2}{5})}{(\frac{2}{5})^2} \\
 &= \frac{5(\frac{3}{5})}{(\frac{4}{25})} \\
 &= \frac{3 \times 25}{4} \\
 &= \frac{125}{4}
 \end{aligned}$$

Section 6.2 Problem Sets

~~8, 10, 12, 14, 16~~ 8, 10, 12, 14, 16

- 8) $\lambda_1 = 5$ accidents/hour $8 \rightarrow 9$ am $\Rightarrow 1$ hr
 $\lambda_2 = 3$ acc/hour $9 \rightarrow 11$ am $\Rightarrow 2$ hr

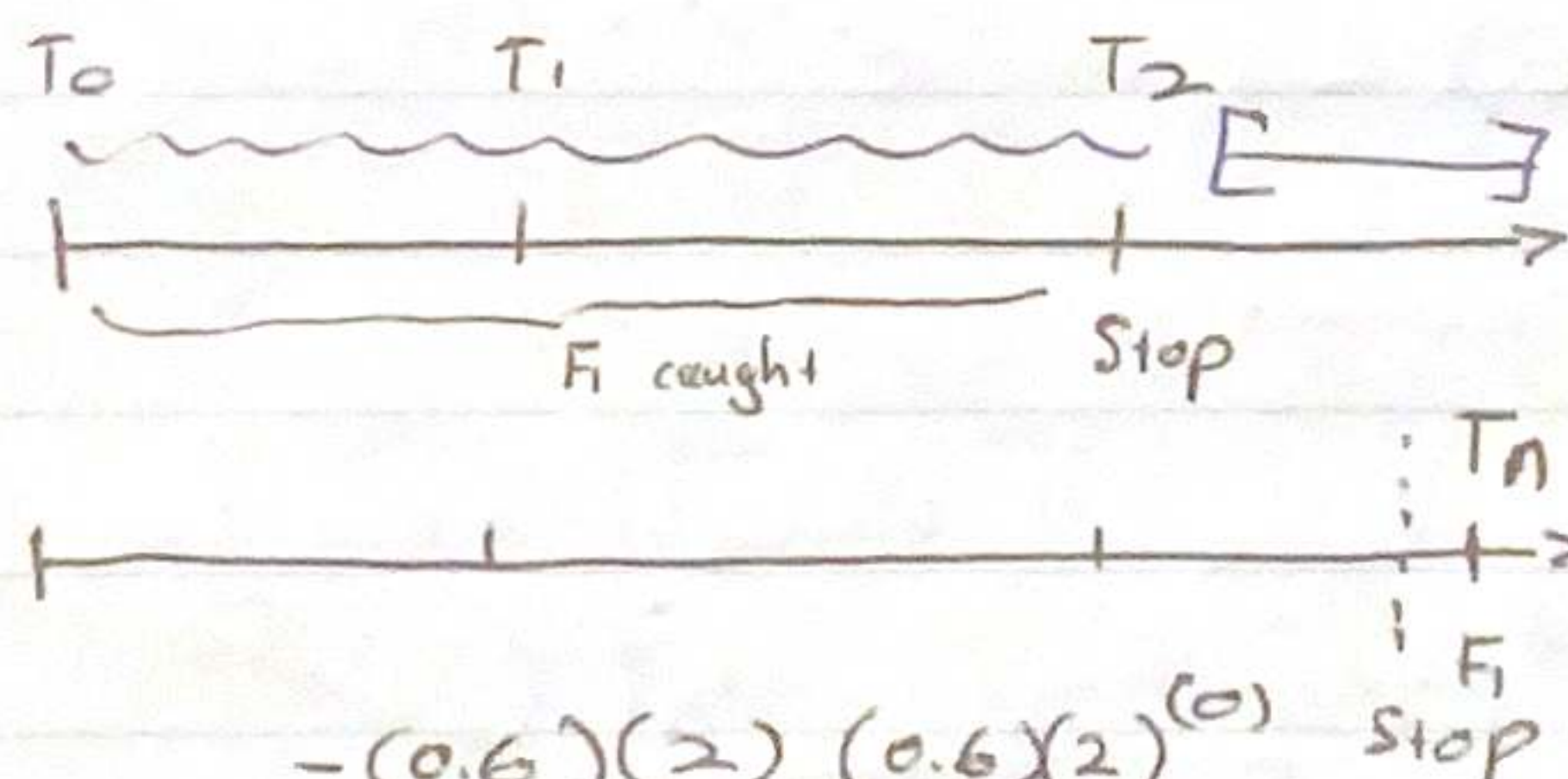
PMF for 8 am $\rightarrow$ 11 am

$$\lambda_c = \lambda_1 + \lambda_2 = 5 + 3 = 8 \text{ accidents/hour.}$$

Poisson PMF with

$$\begin{aligned} \text{PMF (total \# accidents)} &= (1)(5) + (2)(3) \\ &= \underline{11} \text{ accidents between 8 am/11 am} \end{aligned}$$

- 10) $\lambda_1 = 0.6$ per hour



$$\begin{aligned} a) P(0, 2) &= e^{-(0.6)(2)} \cdot \frac{(0.6 \times 2)^{(0)}}{(0)!} \\ &= \cancel{0.3012} \cdot 0.3012 \quad \checkmark \end{aligned}$$

$$\begin{aligned} b) P(2 \text{ to } 5 \text{ hrs}) &= P(1, 5) - P(0, 2) \\ &= \left[e^{-(0.6 \times 5)} \cdot \frac{(0.6 \times 5)^{(1)}}{(1)!} \right] - \left[e^{-(0.6 \times 2)} \cdot \frac{(0.6 \times 2)^{(0)}}{(0)!} \right] \\ &= 0.14936 - 0.3012 \quad \times \end{aligned}$$

$$\begin{aligned} P(2 \text{ to } 5 \text{ hrs}) &= P(0 \text{ to } 2) P(3 \text{ to } 5) \\ &= P(0, 2) (1 - P(0, 3)) \\ &= e^{-(0.6 \times 2)} [1 - e^{-(0.6 \times 3)}] \\ &= \underline{0.2514} \quad \checkmark \end{aligned}$$

$$c) P(\text{Catch} \geq 2 \text{ fish}) = ?$$

$$= P(2, 2) = e^{-(0.6)(2)} \cdot \frac{((0.6)(2))^2}{2!}$$

$$= 0.21685 \times$$

could catch > 2

$$= 1 - P(0, 2) - P(1, 2)$$

$$= 0.337$$

$$d) E[N] = \lambda T = (0.6)T$$

2x Cases $\rightarrow$ Separate into discrete (indep) time intervals.

$$E[N] = E[N_{0 \rightarrow 2}] + E[N_{2 \rightarrow \infty}]$$

$$= \lambda T + E[N_{2 \rightarrow \infty} = 1]$$

$$= (0.6)(2) + P(0, 2)$$

$$= (0.6)(2) + e^{-(0.6)(2)} \cdot 1$$

$$= 1.5012$$

$$e) E[\text{Total Fishing Time} \mid \text{already 4 hrs}] = E[\text{Time after } T_4]$$

$$= E[\text{Time given } T_{\text{current}} = 4]$$

$$= \frac{1}{\lambda} + 4 = \frac{1}{0.6} + 4$$

$$= 1.6667 + 4$$

$$= 5.67 \text{ hours}$$

12) n Pizza types K customers $\rightarrow$ Poisson (λ)

$$X = X_1 + X_2 + X_3 + X_4 + \dots + X_n$$

$$E[X] = n E[X_1] \quad X_i = \begin{cases} 1 & \text{type } i \text{ pizza ordered} \\ 0 & \text{otherwise} \end{cases}$$

$$E[X_1] = P(Y_1 > 0) = 1 - P(Y_1 = 0)$$

$\hookrightarrow$ type 1 ordered at least once $= 1 - e^{-(\lambda)(\frac{1}{n})}$
 $= 1 - e^{-\lambda/n}$

$$E[X] = n E[X_1] = n(1 - e^{-\lambda/n})$$

14) Walkthrough Memo

$$\begin{aligned} a) E[X] &= 0.5 E[T_A] + 0.5 E[T_B] \\ &= 0.5 \left(\frac{1}{\lambda} \right) + 0.5 \left(\frac{1}{3\lambda} \right) \\ &= 0.5(1) + 0.5\left(\frac{1}{3}\right) \\ &= \frac{2}{3} \checkmark \end{aligned}$$

$$f(t) = \lambda e^{-\lambda t}$$

b) ~~$P(C, t) \neq$~~ Let C = no bulb fails before time t .

$$\begin{aligned} P(C) &= P(C|A)P(A) + P(C|B)P(B) \\ &= (e^{-\lambda})(\frac{1}{2}) + (e^{-3\lambda})(\frac{1}{2}) \\ &= \frac{1}{2}e^{-\lambda} + \frac{1}{2}e^{-3\lambda} \checkmark \end{aligned}$$

$$c) P(A|C) = \frac{P(A \cap C)}{P(C)}$$

$$= \frac{\frac{1}{2}e^{-\lambda}}{\frac{1}{2}e^{-\lambda} + \frac{1}{2}e^{-3\lambda}}$$

$$= \frac{1}{1 + e^{-2\lambda}} \checkmark$$

$$\left(\frac{\frac{1}{2}e^{-\lambda}}{\frac{1}{2}e^{-\lambda} + \frac{1}{2}e^{-3\lambda}} \right)$$

$$d) \text{var}(X) = E[X^2] - E[X]^2$$

$$E[X^2] = E[X^2|A]P(A) + E[X^2|B]P(B) \\ = \left(2\right)\left(\frac{1}{2}\right) + \left(\frac{2}{9}\right)\left(\frac{1}{2}\right)$$

$$\rightarrow \left\{ \begin{aligned} E[T^2] &= E[T]^2 + \text{var}(T) \\ &= \left(\frac{1}{\lambda}\right)^2 + \frac{1}{\lambda^2} \\ &= \frac{2}{\lambda^2} \end{aligned} \right\} \begin{array}{l} \text{2nd Moment of an} \\ \text{Exponential RV } T \end{array}$$

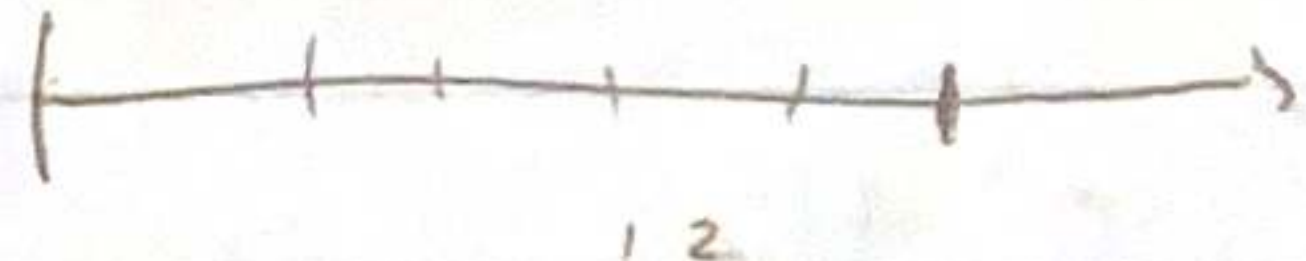
$$= \frac{10}{9}$$

e) 12th bulb failure $\rightarrow$ 4th type A bulb failure.
 $\hookrightarrow$ Let $\sim = E$

~~P(E)~~

$$P(E) = P(\text{first 11 bulb failures have 3 type A}) \times P(12\text{th}) \\ = \binom{11}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^8 \times \left(\frac{1}{2}\right) \\ = 0.00028$$

$T_{12} \Rightarrow A_4$



f) $P(4 \text{ type As Fail in 12 Bulbs})$

$$= \binom{12}{4} \left(\frac{1}{2}\right)^{12}$$

g) ~~$P(T_{12} = t) = \frac{1}{2} (e^{-t} + e^{-3t})$~~ PDF: $\frac{1}{2} (e^{-x} + e^{-3x})$

time ~~let~~ until 12th fail

$\hookrightarrow$ independent

$\hookrightarrow$ 11th fail $\rightarrow$ 12th fail

$\hookrightarrow$ expected time 12